NBA Total Points Prediction: Predictive Modeling

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Library Loading

```r
library(readr)
```

```
## Warning: package 'readr' was built under R version 4.5.2
```

```r
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 4.5.2

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(tidyr)
```

```
## Warning: package 'tidyr' was built under R version 4.5.2
```

```r
library(ggplot2)
library(caret)
```

```
## Warning: package 'caret' was built under R version 4.5.2

## Loading required package: lattice
```

```r
library(glmnet)
```

```
## Warning: package 'glmnet' was built under R version 4.5.2

## Loading required package: Matrix

##
## Attaching package: 'Matrix'

## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack

## Loaded glmnet 4.1-10
```

```r
library(rpart)
library(car)
```

```
## Warning: package 'car' was built under R version 4.5.2

## Loading required package: carData

##
## Attaching package: 'car'

## The following object is masked from 'package:dplyr':
##
##     recode
```

```r
library(rpart.plot)
```

```
## Warning: package 'rpart.plot' was built under R version 4.5.2
```

```r
library(randomForest)
```

```
## Warning: package 'randomForest' was built under R version 4.5.2

## randomForest 4.7-1.2

## Type rfNews() to see new features/changes/bug fixes.

##
## Attaching package: 'randomForest'

## The following object is masked from 'package:ggplot2':
##
##     margin

## The following object is masked from 'package:dplyr':
##
##     combine
```

```r
library(gbm)
```

```
## Warning: package 'gbm' was built under R version 4.5.2

## Loaded gbm 2.2.3

## This version of gbm is no longer under development. Consider transitioning to gbm3, https://github.c
```

```r
library(nnet)
library(Metrics)
```

```
## Warning: package 'Metrics' was built under R version 4.5.2

##
## Attaching package: 'Metrics'

## The following objects are masked from 'package:caret':
##
##     precision, recall
```

```r
library(tibble)
library(lubridate)
```

```
## Warning: package 'lubridate' was built under R version 4.5.2

##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```r
library(purrr)
```

```
## Warning: package 'purrr' was built under R version 4.5.2

##
## Attaching package: 'purrr'

## The following object is masked from 'package:car':
##
##     some

## The following object is masked from 'package:caret':
##
##     lift
```

```r
library(e1071)
```

```
## Warning: package 'e1071' was built under R version 4.5.2

##
## Attaching package: 'e1071'

## The following object is masked from 'package:ggplot2':
##
##     element
```

```r
library(rprojroot)
```

```
## Warning: package 'rprojroot' was built under R version 4.5.2
```

```r
library(corrplot)
```

```
## Warning: package 'corrplot' was built under R version 4.5.2

## corrplot 0.95 loaded
```

```r
model_ready <- read_csv("Data/FINAL_MODEL_READY.csv", show_col_types = FALSE) |>
  mutate(game_date = as.Date(game_date)) |>
  filter(
    game_date >= as.Date("2013-10-01"),
    month(game_date) %in% c(10, 11, 12, 1, 2, 3, 4)
  )

cat("Total rows:", nrow(model_ready), "\n")
```

```
## Total rows: 15702
```

```r
cat("Date range:", as.character(min(model_ready$game_date)),
    "to", as.character(max(model_ready$game_date)), "\n")
```

```
## Date range: 2013-10-18 to 2026-03-03
```

TP Dist.

```r
summary(model_ready$total)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   134.0   202.0   217.0   217.5   233.0   351.0
```

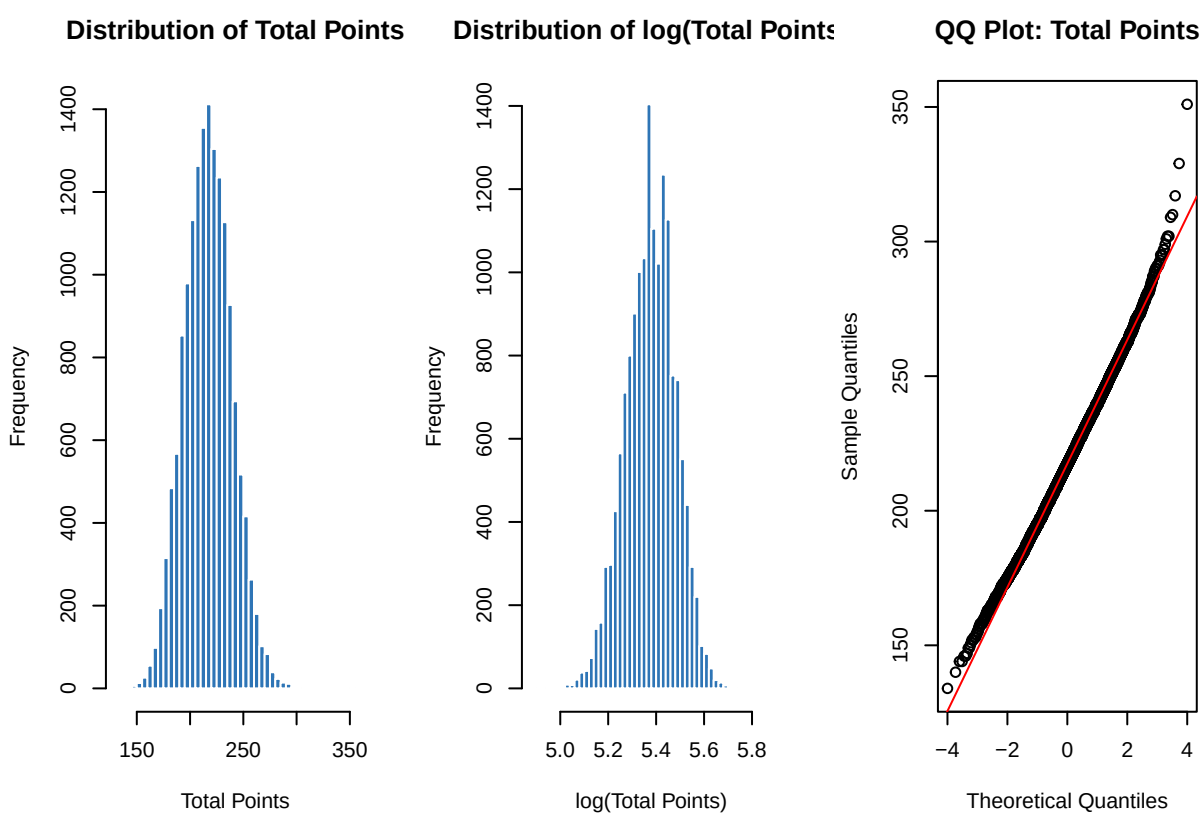
```r
cat("Skewness:", round(e1071::skewness(model_ready$total), 3), "\n")
```

```
## Skewness: 0.139
```

```r
cat("Kurtosis:", round(e1071::kurtosis(model_ready$total), 3), "\n")
```

```
## Kurtosis: 0.088
```

```r
par(mfrow = c(1, 3))

hist(model_ready$total,
     breaks = 50,
     main   = "Distribution of Total Points",
     xlab   = "Total Points",
     col    = "#2E75B6", border = "white")

hist(log(model_ready$total),
     breaks = 50,
     main   = "Distribution of log(Total Points)",
     xlab   = "log(Total Points)",
     col    = "#2E75B6", border = "white")

qqnorm(model_ready$total, main = "QQ Plot: Total Points")
qqline(model_ready$total, col = "red")
```



```r
par(mfrow = c(1, 1))
```

Train and testing data

```r
cutoff_date <- as.Date("2025-03-13")

train <- model_ready |> filter(game_date <  cutoff_date)
test  <- model_ready|> filter(game_date >= cutoff_date)

cat("Training set:", nrow(train), "games (",
    as.character(min(train$game_date)), "to",
    as.character(max(train$game_date)), ")\n")
```

```
## Training set: 14493 games ( 2013-10-18 to 2025-03-12 )
```

```r
cat("Test set:    ", nrow(test),  "games (",
    as.character(min(test$game_date)),  "to",
    as.character(max(test$game_date)),  ")\n")
```

```
## Test set:     1209 games ( 2025-03-13 to 2026-03-03 )
```

Feature matrix

```r
id_cols <- c("game_id", "game_date", "home_team", "away_team", "total", "oreb")

numeric_data <- model_ready |>
  select(where(is.numeric)) |>
  select(-any_of(c("game_id", "total", "oreb")))

# remove zero-variance columns first
zero_var <- names(numeric_data)[sapply(
  numeric_data,
  function(x) var(x, na.rm = TRUE) == 0 | all(is.na(x))
)]

if (length(zero_var) > 0) {
  cat("Dropping zero-variance columns:\n")
  print(zero_var)
  numeric_data <- numeric_data |>
    select(-all_of(zero_var))
}

# remove obvious collinear / redundant variables before VIF
drop_for_vif <- c(
  "blend_total_proj5",
  "rebound_matchup_edge",
  "orb_pg_diff",
  "drb_pg_diff",
```

```r
  "trb_pg_diff",
  "x3pa_rate_pg_diff",
  "shooting_env5",
  "pace_pressure5",
  "free_throw_env5",
  "tov_env5",
  "pace_rebound_pressure"
)

vif_vars <- setdiff(names(numeric_data), drop_for_vif)

lm_test <- lm(
  total ~ .,
  data = model_ready |>
    select(total, all_of(vif_vars))
)

aliased <- alias(lm_test)$Complete
if (!is.null(aliased)) {
  cat("Aliased coefficients detected. Remove some of these variables:\n")
  print(aliased)
}

keep_coef <- names(coef(lm_test))[!is.na(coef(lm_test))]
keep_coef <- setdiff(keep_coef, "(Intercept)")

lm_test_vif <- lm(
  as.formula(paste("total ~", paste(keep_coef, collapse = " + "))),
  data = model_ready |>
    select(total, all_of(keep_coef))
)

cat("VIF values:\n")
```

```
## VIF values:
```

```r
print(car::vif(lm_test_vif))
```

```
##                   spread     opp_pts_allowed_diff     opp_orb_allowed_diff
##                 1.227101                 4.361487                 4.295419
##     opp_trb_allowed_diff     opp_ast_allowed_diff      opp_tov_forced_diff
##                13.234133                 2.037228                 2.371881
##           last5_pts_diff   last5_pts_allowed_diff          last5_oreb_diff
##                 1.708824                 1.698755                 3.930409
##                rest_diff        home_back_to_back        away_back_to_back
##                 1.038664                 1.043948                 1.052558
##        missed_fg_pg_diff       expected_missed_fg          orb_per100_diff
##                13.500952                 1.337574               587.265348
##          drb_per100_diff          trb_per100_diff    x3pa_rate_per100_diff
##               758.439877              1276.204376                 3.973846
##     last5_missed_fg_diff    last10_missed_fg_diff    last5_missed_3pt_diff
##                 6.978400                 8.375342                20.075272
##   last10_missed_3pt_diff     last5_x3pa_rate_diff    last10_x3pa_rate_diff
```

```
##                30.150491                20.325298                31.388908
##         last10_oreb_diff  last5_oreb_allowed_diff last10_oreb_allowed_diff
##                 4.910462                64.425623                 2.653055
##   expected_oreb_pressure        home_oreb_matchup        away_oreb_matchup
##                 1.959680                54.620162                54.113473
```

```r
# final predictor set for modeling
feat_cols <- setdiff(names(model_ready), id_cols)

# drop collinear / redundant vars from actual models too
feat_drop <- c(
  "blend_total_proj5",
  "rebound_matchup_edge",
  "orb_pg_diff",
  "drb_pg_diff",
  "trb_pg_diff",
  "x3pa_rate_pg_diff",
  "shooting_env5",
  "pace_pressure5",
  "free_throw_env5",
  "tov_env5",
  "pace_rebound_pressure"
)

feat_cols <- setdiff(feat_cols, feat_drop)

cat("Number of predictors:", length(feat_cols), "\n")
```

```
## Number of predictors: 30
```

```r
cat("Predictors:\n")
```

```
## Predictors:
```

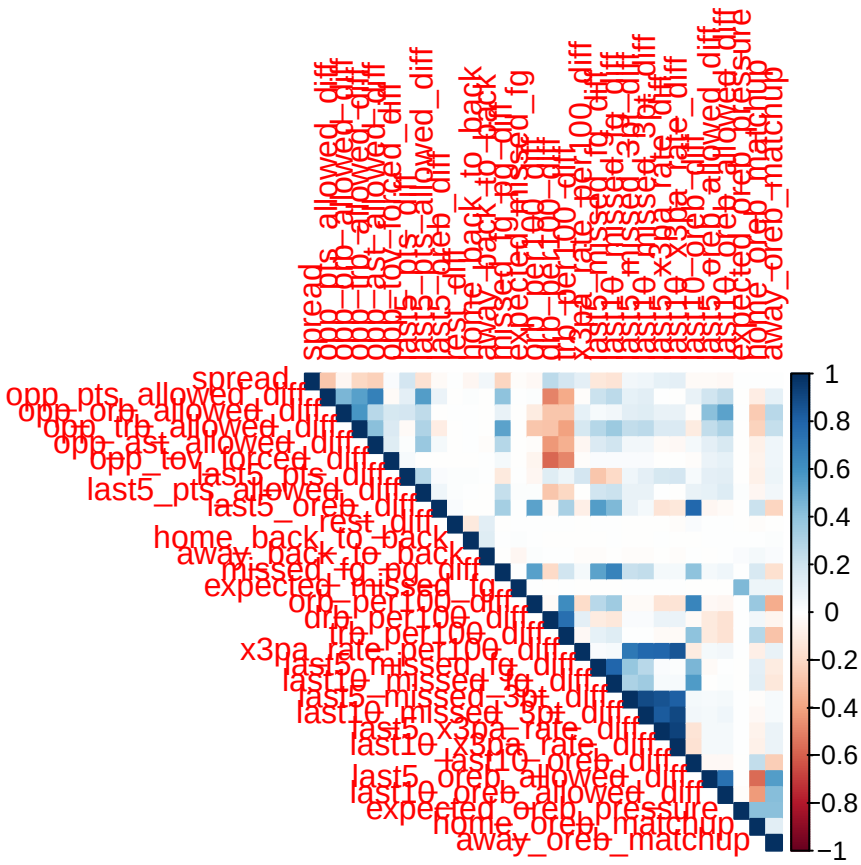
```r
print(feat_cols)
```

```
##  [1] "spread"                   "opp_pts_allowed_diff"
##  [3] "opp_orb_allowed_diff"     "opp_trb_allowed_diff"
##  [5] "opp_ast_allowed_diff"     "opp_tov_forced_diff"
##  [7] "last5_pts_diff"           "last5_pts_allowed_diff"
##  [9] "last5_oreb_diff"          "rest_diff"
## [11] "home_back_to_back"        "away_back_to_back"
## [13] "missed_fg_pg_diff"        "expected_missed_fg"
## [15] "orb_per100_diff"          "drb_per100_diff"
## [17] "trb_per100_diff"          "x3pa_rate_per100_diff"
## [19] "last5_missed_fg_diff"     "last10_missed_fg_diff"
## [21] "last5_missed_3pt_diff"    "last10_missed_3pt_diff"
## [23] "last5_x3pa_rate_diff"     "last10_x3pa_rate_diff"
## [25] "last10_oreb_diff"         "last5_oreb_allowed_diff"
## [27] "last10_oreb_allowed_diff" "expected_oreb_pressure"
## [29] "home_oreb_matchup"        "away_oreb_matchup"
```

```r
X_train <- train[, feat_cols, drop = FALSE]
y_train <- train$total

X_test  <- test[, feat_cols, drop = FALSE]
y_test  <- test$total

y_train_log <- log(y_train)
y_test_log  <- log(y_test)

na_counts <- colSums(is.na(X_train))
if (any(na_counts > 0)) {
  cat("WARNING - NAs in training features:\n")
  print(na_counts[na_counts > 0])
} else {
  cat("No NAs in training feature matrix.\n")
}
```

```
## WARNING - NAs in training features:
##     last5_missed_fg_diff    last10_missed_fg_diff    last5_missed_3pt_diff
##                        5                       10                        5
##   last10_missed_3pt_diff     last5_x3pa_rate_diff    last10_x3pa_rate_diff
##                       10                        5                       10
##         last10_oreb_diff  last5_oreb_allowed_diff last10_oreb_allowed_diff
##                       10                        5                       10
##   expected_oreb_pressure        home_oreb_matchup        away_oreb_matchup
##                        5                        2                        3
```

```r
cor_matrix <- cor(X_train |> select(where(is.numeric)), use = "complete.obs")
corrplot(cor_matrix, method = "color", type = "upper")
```




```r
cor_with_total <- sapply(
  X_train |> select(where(is.numeric)),
  function(x) cor(y_train, x, use = "complete.obs")
)

cat("Top correlations with total:\n")
```

```
## Top correlations with total:
```

```r
print(sort(abs(cor_with_total), decreasing = TRUE))
```

```
##       expected_missed_fg   expected_oreb_pressure        away_back_to_back
##             2.064992e-01             5.255727e-02             5.107155e-02
##        away_oreb_matchup        home_oreb_matchup     last5_missed_fg_diff
##             3.577479e-02             2.877893e-02             2.176230e-02
##     opp_orb_allowed_diff                   spread        missed_fg_pg_diff
##             1.346353e-02             1.279984e-02             1.265302e-02
##  last5_oreb_allowed_diff    last10_missed_fg_diff     opp_trb_allowed_diff
##             1.212023e-02             1.053013e-02             9.797702e-03
## last10_oreb_allowed_diff          orb_per100_diff         last10_oreb_diff
##             9.211636e-03             7.572339e-03             7.129787e-03
##          drb_per100_diff      opp_tov_forced_diff    last5_missed_3pt_diff
##             6.669519e-03             6.591693e-03             6.386571e-03
##          last5_oreb_diff                 rest_diff     opp_ast_allowed_diff
```

```
##             5.808712e-03             3.300975e-03             3.264590e-03
##     opp_pts_allowed_diff   last10_missed_3pt_diff        home_back_to_back
##             2.863070e-03             2.851603e-03             1.526077e-03
##    last10_x3pa_rate_diff           last5_pts_diff   last5_pts_allowed_diff
##             1.453639e-03             5.881399e-04             4.282906e-04
##     last5_x3pa_rate_diff    x3pa_rate_per100_diff          trb_per100_diff
##             3.130274e-04             2.457291e-04             9.684296e-05
```

Cross-validation

```r
ts_ctrl <- trainControl(
  method          = "timeslice",
  initialWindow   = floor(nrow(train) * 0.6),
  horizon         = floor(nrow(train) * 0.08),
  fixedWindow     = FALSE,
  summaryFunction = defaultSummary,
  allowParallel   = TRUE
)

cv5_ctrl <- trainControl(
  method   = "cv",
  number   = 5,
  summaryFunction = defaultSummary
)

mae_raw <- function(actual, predicted) {
  mean(abs(actual - predicted), na.rm = TRUE)
}

mae_log_backtransform <- function(actual, pred_log) {
  mean(abs(actual - exp(pred_log)))
}
```

Baseline Model

```r
baseline_pred <- rep(mean(y_train), length(y_test))
baseline_mae  <- mae_raw(y_test, baseline_pred)

cat("Baseline (training mean prediction):", round(mean(y_train), 2), "points\n")
```

```
## Baseline (training mean prediction): 216.53 points
```

```r
cat("Baseline MAE on test set:", round(baseline_mae, 2), "\n")
```

```
## Baseline MAE on test set: 18.84
```

Model 1: Ordinary Least Squares Linear Regression

o l s l o l

Full Model 1

```r
lm_full <- lm(total ~ ., data = cbind(total = y_train, X_train))
lm_full_pred <- predict(lm_full, newdata = X_test)
lm_full_mae <- mae_raw(y_test, lm_full_pred)

cat("OLS Full Model - Test MAE:", round(lm_full_mae, 2), "\n")
```

```
## OLS Full Model - Test MAE: 18.74
```

```r
cat("R-squared (training):", round(summary(lm_full)$r.squared, 4), "\n")
```

```
## R-squared (training): 0.0729
```

```r
if (requireNamespace("car", quietly = TRUE)) {
  vif_vals <- car::vif(lm_full)
  cat("\nPredictors with VIF > 10 (multicollinearity concern):\n")
  print(vif_vals[vif_vals > 10])
}
```

```
##
## Predictors with VIF > 10 (multicollinearity concern):
##    opp_trb_allowed_diff       missed_fg_pg_diff         orb_per100_diff
##                13.10012                13.56558               560.34507
##         drb_per100_diff         trb_per100_diff   last5_missed_3pt_diff
##               757.70277              1244.54927                20.30735
##  last10_missed_3pt_diff    last5_x3pa_rate_diff   last10_x3pa_rate_diff
##                30.33935                20.68380                31.75806
## last5_oreb_allowed_diff       home_oreb_matchup       away_oreb_matchup
##                68.50055                56.42244                55.93446
```

Pruned OLS Model 1

```r
pruned_vars <- c(
  "spread",
  "rest_diff",
  "home_back_to_back",
  "away_back_to_back",
  "last5_pts_diff",
  "last5_pts_allowed_diff",
  "last5_oreb_diff",
  "opp_pts_allowed_diff",
  "opp_orb_allowed_diff",
  "opp_trb_allowed_diff",
  "opp_ast_allowed_diff",
  "opp_tov_forced_diff",
  "expected_oreb_pressure",
  "last5_missed_fg_diff",
  "last10_oreb_diff",
```

```r
  "last5_oreb_allowed_diff",
  "orb_per100_diff",
  "drb_per100_diff",
  "trb_per100_diff",
  "x3pa_rate_per100_diff"
)

pruned_vars <- pruned_vars[pruned_vars %in% names(X_train)]

lm_pruned <- lm(
  total ~ .,
  data = cbind(total = y_train, X_train[, pruned_vars, drop = FALSE])
)

lm_pruned_pred <- predict(lm_pruned, newdata = X_test[, pruned_vars, drop = FALSE])
lm_pruned_mae  <- mae_raw(y_test, lm_pruned_pred)

cat("OLS Pruned Model - Test MAE:", round(lm_pruned_mae, 2), "\n")
```

```
## OLS Pruned Model - Test MAE: 19.23
```

```r
cat("Pruned R-squared (training):", round(summary(lm_pruned)$r.squared, 4), "\n")
```

```
## Pruned R-squared (training): 0.0066
```

```r
print(summary(lm_pruned))
```

```
##
## Call:
## lm(formula = total ~ ., data = cbind(total = y_train, X_train[,
##     pruned_vars, drop = FALSE]))
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -81.13 -15.42  -0.26  14.78 135.64
##
## Coefficients:
##                          Estimate Std. Error t value Pr(>|t|)
## (Intercept)             227.31449    1.65785 137.114  < 2e-16 ***
## spread                   -0.02039    0.01427  -1.428  0.15320
## rest_diff                 0.02057    0.04884   0.421  0.67356
## home_back_to_back         0.55638    0.54731   1.017  0.30938
## away_back_to_back        -2.70164    0.45029  -6.000 2.02e-09 ***
## last5_pts_diff           -0.03609    0.02590  -1.393  0.16354
## last5_pts_allowed_diff    0.03006    0.02574   1.168  0.24284
## last5_oreb_diff           0.15795    0.11606   1.361  0.17356
## opp_pts_allowed_diff      0.05822    0.06864   0.848  0.39635
## opp_orb_allowed_diff     -0.25814    0.24147  -1.069  0.28508
## opp_trb_allowed_diff      0.01266    0.12372   0.102  0.91849
## opp_ast_allowed_diff     -0.05804    0.12669  -0.458  0.64686
## opp_tov_forced_diff       0.10897    0.16546   0.659  0.51015
## expected_oreb_pressure   -1.02468    0.16563  -6.187 6.31e-10 ***
```

```
## last5_missed_fg_diff     -0.18513    0.05717  -3.238  0.00121 **
## last10_oreb_diff         -0.02717    0.14361  -0.189  0.84994
## last5_oreb_allowed_diff  -0.07899    0.08047  -0.982  0.32633
## orb_per100_diff           1.45555    2.55721   0.569  0.56923
## drb_per100_diff           1.71735    2.56570   0.669  0.50328
## trb_per100_diff          -1.52899    2.55533  -0.598  0.54961
## x3pa_rate_per100_diff     1.70041    3.23130   0.526  0.59874
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22.2 on 14462 degrees of freedom
##   (10 observations deleted due to missingness)
## Multiple R-squared:  0.006607,   Adjusted R-squared:  0.005233
## F-statistic: 4.809 on 20 and 14462 DF,  p-value: 6.691e-12
```

OLS on Log-Transformed Total Model 1

```r
lm_log <- lm(
  log(total) ~ .,
  data = cbind(total = y_train, X_train[, pruned_vars, drop = FALSE])
)

lm_log_pred_log <- predict(lm_log, newdata = X_test[, pruned_vars, drop = FALSE])
lm_log_pred_raw <- exp(lm_log_pred_log)
lm_log_mae      <- mae_raw(y_test, lm_log_pred_raw)

cat("OLS Log-Transformed - Test MAE (back-transformed):", round(lm_log_mae, 2), "\n")
```

```
## OLS Log-Transformed - Test MAE (back-transformed): 19.86
```

Model 2: Polynomial Regression

```r
poly_base_vars <- intersect(
  c(
    "spread",
    "rest_diff",
    "last5_pts_diff",
    "last5_pts_allowed_diff",
    "last5_oreb_diff",
    "opp_pts_allowed_diff",
    "opp_orb_allowed_diff",
    "expected_oreb_pressure",
    "last5_missed_fg_diff",
    "last5_oreb_allowed_diff"
  ),
  names(X_train)
)

poly_terms <- intersect(
  c("expected_oreb_pressure", "last5_pts_diff", "last5_pts_allowed_diff"),
```

```r
  names(X_train)
)

train_poly <- cbind(total = y_train, X_train[, poly_base_vars, drop = FALSE])
test_poly  <- X_test[, poly_base_vars, drop = FALSE]

if ("expected_oreb_pressure" %in% poly_terms) {
  train_poly <- cbind(
    train_poly,
    poly(X_train$expected_oreb_pressure, 2, raw = TRUE) |>
      as.data.frame() |> setNames(c("orebpress_d1", "orebpress_d2"))
  )
  test_poly <- cbind(
    test_poly,
    poly(X_test$expected_oreb_pressure, 2, raw = TRUE) |>
      as.data.frame() |> setNames(c("orebpress_d1", "orebpress_d2"))
  )
}

if ("last5_pts_diff" %in% poly_terms) {
  train_poly <- cbind(
    train_poly,
    poly(X_train$last5_pts_diff, 2, raw = TRUE) |>
      as.data.frame() |> setNames(c("last5pts_d1", "last5pts_d2"))
  )
  test_poly <- cbind(
    test_poly,
    poly(X_test$last5_pts_diff, 2, raw = TRUE) |>
      as.data.frame() |> setNames(c("last5pts_d1", "last5pts_d2"))
  )
}

if ("last5_pts_allowed_diff" %in% poly_terms) {
  train_poly <- cbind(
    train_poly,
    poly(X_train$last5_pts_allowed_diff, 2, raw = TRUE) |>
      as.data.frame() |> setNames(c("last5allow_d1", "last5allow_d2"))
  )
  test_poly <- cbind(
    test_poly,
    poly(X_test$last5_pts_allowed_diff, 2, raw = TRUE) |>
      as.data.frame() |> setNames(c("last5allow_d1", "last5allow_d2"))
  )
}

lm_poly <- lm(total ~ ., data = train_poly)
lm_poly_pred <- predict(lm_poly, newdata = test_poly)
lm_poly_mae  <- mae_raw(y_test, lm_poly_pred)

cat("Polynomial Regression (degree 2) - Test MAE:", round(lm_poly_mae, 2), "\n")
```

```
## Polynomial Regression (degree 2) - Test MAE: 19.3
```

```r
cat("Polynomial R-squared (training):", round(summary(lm_poly)$r.squared, 4), "\n")
```

```
## Polynomial R-squared (training): 0.0039
```

```r
poly_coefs <- summary(lm_poly)$coefficients
cat("\nSquared-term significance (p-values):\n")
```

```
##
## Squared-term significance (p-values):
```

```r
print(round(poly_coefs[grep("_d2", rownames(poly_coefs)), 4], 4))
```

```
##  orebpress_d2   last5pts_d2 last5allow_d2
##        0.8347        0.8130        0.8427
```

Model 3: Gradient Boosted Trees (GBM)

```r
gbm_grid <- expand.grid(
  n.trees           = c(200, 500),
  interaction.depth = c(2, 4),
  shrinkage         = 0.1,
  n.minobsinnode    = 20
)

set.seed(42)
gbm_tune <- train(
  x         = as.data.frame(X_train),
  y         = y_train,
  method    = "gbm",
  metric    = "MAE",
  trControl = trainControl(method = "cv", number = 3, verboseIter = FALSE),
  tuneGrid  = gbm_grid,
  verbose   = FALSE
)

best_gbm_params <- gbm_tune$bestTune
cat("=== Best GBM Hyperparameters (3-fold CV) ===\n")
```

```
## === Best GBM Hyperparameters (3-fold CV) ===
```

```r
print(best_gbm_params)
```

```
##   n.trees interaction.depth shrinkage n.minobsinnode
## 3     200                 4       0.1             20
```

```r
# Refit on full training set (raw total)
set.seed(42)
gbm_train_df <- as.data.frame(cbind(total = y_train, X_train))
gbm_test_df  <- as.data.frame(X_test)

gbm_final <- gbm(
  formula           = total ~ .,
  data              = gbm_train_df,
  distribution      = "gaussian",
  n.trees           = best_gbm_params$n.trees,
  interaction.depth = best_gbm_params$interaction.depth,
  shrinkage         = best_gbm_params$shrinkage,
  n.minobsinnode    = best_gbm_params$n.minobsinnode,
  bag.fraction      = 0.75,
  train.fraction    = 1.0,
  verbose           = FALSE
)

gbm_pred <- predict(
  gbm_final,
  newdata = gbm_test_df,
  n.trees = best_gbm_params$n.trees
)

gbm_mae <- mae_raw(y_test, gbm_pred)
cat("GBM Final Model - Test MAE:", round(gbm_mae, 2), "\n")
```

```
## GBM Final Model - Test MAE: 18.31
```

```r
# Variable importance
cat("\nGBM - Variable Importance (top 15):\n")
```

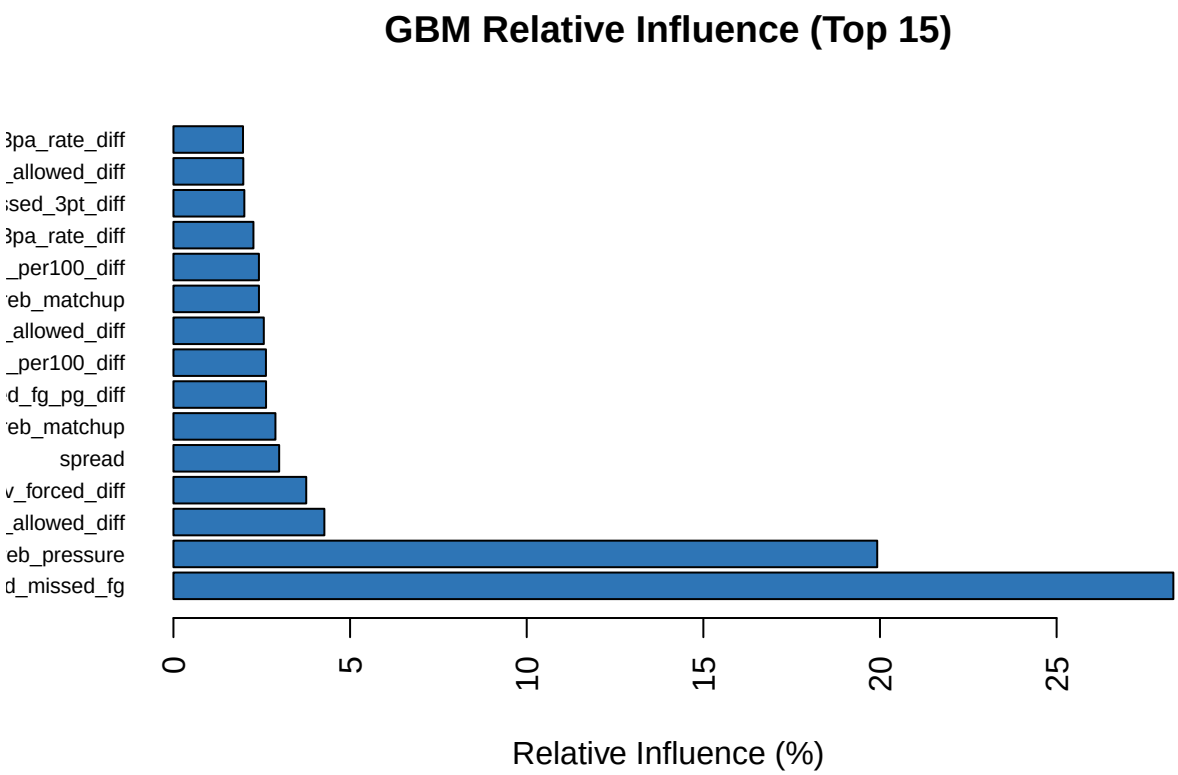
```
##
## GBM - Variable Importance (top 15):
```

```r
gbm_summary <- summary(
  gbm_final,
  n.trees = best_gbm_params$n.trees,
  plotit = FALSE
)
print(head(gbm_summary, 15))
```

```
##                                                var   rel.inf
## expected_missed_fg             expected_missed_fg 28.304464
## expected_oreb_pressure     expected_oreb_pressure 19.921671
## opp_ast_allowed_diff         opp_ast_allowed_diff  4.272418
## opp_tov_forced_diff           opp_tov_forced_diff  3.758832
## spread                                     spread  2.992895
## home_oreb_matchup               home_oreb_matchup  2.886977
## missed_fg_pg_diff               missed_fg_pg_diff  2.621886
## drb_per100_diff                   drb_per100_diff  2.618868
## opp_orb_allowed_diff         opp_orb_allowed_diff  2.557977
```

```
## away_oreb_matchup               away_oreb_matchup  2.420713
## trb_per100_diff                   trb_per100_diff  2.419663
## last10_x3pa_rate_diff       last10_x3pa_rate_diff  2.264605
## last10_missed_3pt_diff     last10_missed_3pt_diff  2.006803
## last10_oreb_allowed_diff last10_oreb_allowed_diff  1.977759
## last5_x3pa_rate_diff         last5_x3pa_rate_diff  1.969766
```

```
barplot(
  head(gbm_summary$rel.inf, 15),
  names.arg = head(gbm_summary$var, 15),
  horiz     = TRUE,
  las       = 2,
  main      = "GBM Relative Influence (Top 15)",
  xlab      = "Relative Influence (%)",
  col       = "#2E75B6",
  cex.names = 0.7
)
```




```
# Log-scale GBM
set.seed(42)
gbm_log_train_df <- as.data.frame(cbind(total_log = y_train_log, X_train))
gbm_log_test_df  <- as.data.frame(X_test)

gbm_log_final <- gbm(
  formula           = total_log ~ .,
```

```r
  data              = gbm_log_train_df,
  distribution      = "gaussian",
  n.trees           = best_gbm_params$n.trees,
  interaction.depth = best_gbm_params$interaction.depth,
  shrinkage         = best_gbm_params$shrinkage,
  n.minobsinnode    = best_gbm_params$n.minobsinnode,
  bag.fraction      = 0.75,
  train.fraction    = 1.0,
  verbose           = FALSE
)

gbm_log_pred <- exp(
  predict(
    gbm_log_final,
    newdata = gbm_log_test_df,
    n.trees = best_gbm_params$n.trees
  )
)

gbm_log_mae <- mae_raw(y_test, gbm_log_pred)
cat("GBM (log scale, back-transformed) - Test MAE:", round(gbm_log_mae, 2), "\n")
```

```
## GBM (log scale, back-transformed) - Test MAE: 18.86
```